

ASSIGNMENT 01

Name :- Lekkala Dattu Kumar

Reg No :- 192824141

Department :- AI4DS

Course Code :- ITA1412

Course Name :- Ethical Hacking

Case Study 1 :- Unauthorized network scanning Using Nmap.

Scenario :-

A final year engineering student scans a software company's network using Nmap without permission to identify open ports and running services.

Ethical Analysis :-

- * The student intended to learn cybersecurity.
- * However, scanning without authorization is unethical.
- * Ethical hacking requires written permission from the system owner.

Legal Analysis (IT Act 2008)

Applicable Sections :-

- * Section 43 - Unauthorized access or damage to computer system.
- * Section 66 - Computer related offences done when dishonestly or fraudulently.

Possible legal Consequences

- * Civil compensation.
- * Criminal Investigation
- * Fine or imprisonment depending on the offences.
- * Damage to professional reputation.

Correct Procedure :-

- * Obtain return permission.
- * Define testing scope.
- * Sign authorization documents.
- * Perform testing only within approved limits.
- * Submit a responsible reputation.

Conclusion :-

Scanning a network without permission is both the unethical and may be illegal, even if no damage occurs.

Case Study 2 :- Public disclosure of SQL injection vulnerability.

Scenario :-

A researcher discovers a SQL injection vulnerability but immediately post it on social media before of information the company.

Ethical Analysis :-

- * Public disclosure before a fix is irresponsible.
- * It increases the risk of cyberattacks.
- * Researchers should protect users.

Responsible Vulnerability Disclosure.

Responsible Disclosure means :-

1. Notify the organization privately.

2. Gives sufficient time to fix the issue.
3. Publish details only after the patch is released.

Risk of Premature disclosure.

- * Customer data theft.
- * Financial losses.
- * Reputation damage.
- * Legal Consequences.
- * Increased Cybercrime.

Legal Perspective:-

Possible IT Act Scenarios:-

- * Section 43
- * Section 66

If disclosure knowingly facilitates cybercrime, legal action may be taken.

Recommended Steps:-

- * Contact the organizations security team.
- * Share technical evidence privately.
- * Wait for remediation.
- * Publish research only after the vulnerability is fixed.

Conclusion:-

Responsible disclosure protects both organizations and users.

Case Study 3 :- Phishing attack on Financial Organization

Scenario :-

An employee receives a fake email pretending to be IT support and enters login credentials.

Type of Cyberattack :-

Phishing Attack.

- * Social Engineering.
- * Credential Theft.

Legal Implications (IT Act)

Applicable Scenarios :-

- * Section 43 - Unauthorized access.
- * Section 66 - Computer related offences.
- * Section 66C - Identity Theft.
- * Section 66D - Cheating by personation using computer resources.

Responsibilities :-

Organization :-

- * Employee awareness training.
- * Email Security.
- * Multi Factor Authentication
- * Regular Security audits.

Employee :-

- * Verify suspicious emails.

- * Never share passwords.
- * Report phishing immediately.
- * Follow cybersecurity policies.

Organizational:-

- * Cybersecurity awareness training.
- * Phishing simulation.
- * Password policy.
- * Incident response plan.

Conclusion:-

Both employees and organizations share responsibility for preventing phishing attacks.

Case Study 4:- Student Exploits Banking Vulnerability.

Scenario:-

A cybersecurity student finds a banking vulnerability and retrieves a small amount of customer data before operating it.

Ethical Analysis:-

- * Greed intentions does not justify unauthorized exploration.
- * Accessing customer data violates privacy.
- * Ethical hackers must avoid accessing sensitive information.

Ethical Hacking vs Unauthorized Access

Permission obtained.

Legal

Improves Security

Controlled Testing

Applicable IT Act Sections :-

- * Section 43

- * Section 66

- * Section 72.

Proper Reporting Process:-

- * Do not exploit the vulnerability.

- * Collect minimal evidence

- * Inform the bank privately.

- * Request acknowledgement.

- * Wait for the organization to fix the issue.

- * Publish only after permission or remediation.

Conclusion:-

Even with good intentions, unauthorized exploration of vulnerability can lead to legal action.